PID Controller Optimization using Genetic Algorithm (GA)

This project optimizes the PID controller gains for a 2-DOF planar robotic manipulator using Genetic Algorithm in MATLAB/Simulink.

Files

- fitnessPID.m – Fitness function (calculates cost using IAE + torque penalty)
- runGA.m – Script to run GA optimization
- PID_GA.slx – Simulink model of the robot and PID controller

How to Run

1. Set initial PID gains manually (e.g., in untitled2.m or MATLAB workspace).
2. Open MATLAB, set the folder as the working directory.
3. Run the Simulink model PID_GA.slx to confirm it works with the initial gains.
4. Run the optimization script:
   runGA
5. The optimized gains and convergence plot will be displayed in the MATLAB command window.

Disclaimer

When running the GA, the optimized PID gains may differ for each run due to the stochastic nature of the algorithm. However, every set of gains produced by GA is optimized and consistently provides better performance compared to the initial manual gains.

Requirements

MATLAB with Simulink and Global Optimization Toolbox